

Beyond Efficiency

Decision-making in Transport & Mobility

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Up to now, the methods we studied had a strong emphasis on maximizing efficiency.

- Classical problems, such as routing and scheduling, can be solved to near-optimality.

However, there are many other considerations that we would like to incorporate when solving real-life problems:

- Fairness
- Uncertainty
- Multi-agent optimization

The goals of this last module are as follows:

- Introduce key concepts in fair, robust, and multi-agent optimization
- Show that many of our existing solution methods translate to these new settings
- Discuss real-life applications
- Apply key concepts and methods in the group assignment

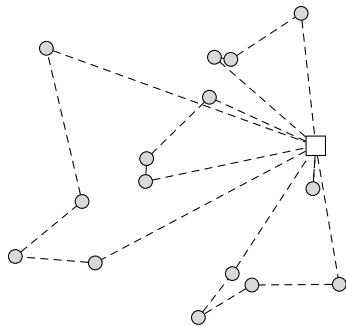
Fairness

Solutions impact perceived work quality of workers, crew members, and so on.

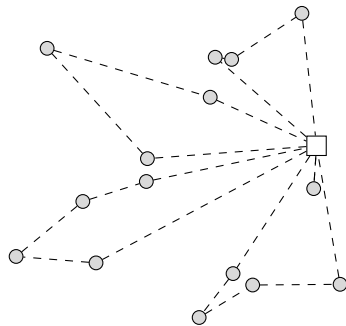
Efficient solutions are not necessarily fair.

By formulating fairness explicitly as objective or constraint, we can improve quality of work at the cost of a small loss in efficiency.

This motivates developing models that balance efficiency with fairness in a principled, quantifiable way.



(a) Total distance: 6,178. Range: 2,170.



(b) Total distance: 6,181. Range: 1,601.

Figure: Efficient (left) and fair (right) vehicle routing solutions.

The fairness question naturally arises in many transport problems:

Application	Agents	Payoff
CVRP	Drivers	Distance, driving time
Crew scheduling	Crew	Track variation, duty length
Garbage collection	Inhabitants	Time until pick-up
Ambulance location	Inhabitants	Response time
Line planning	Travelers	Travel time

Denote by $\mathbf{x} \in \mathcal{X} = \{\mathbf{x} \in \{0, 1\}^k : A\mathbf{x} \geq \mathbf{b}\}$ the set of feasible solutions, e.g., the set of all feasible route combinations. Denote by $\mathbf{c}^\top \mathbf{x}$ the cost of solution $\mathbf{x}$.

We assume there are n agents, e.g., drivers.

Every solution $\mathbf{x}$ maps to a payoff vector $\mathbf{p}(\mathbf{x}) \in \mathbb{R}^n$. Entry p_i is the payoff of agent i , e.g., total workload.

We are interested in minimizing some (un)fairness measure $\phi(\mathbf{p}(\mathbf{x}))$.

Commonly used fairness measures include:

- Min-max: $\min_i \max_i p_i$
- Lexicographic min-max
- Range: $\max_i p_i - \min_i p_i$
- Mean absolute deviation: $\frac{1}{n} \sum_{i=1}^n |p_i - \bar{p}|$
- Standard deviation: $\sqrt{\frac{1}{n} \sum_{i=1}^n (p_i - \bar{p})^2}$
- Gini coefficient: $\frac{1}{2n^2\bar{p}} \sum_{i=1}^n \sum_{j=1}^n |p_i - p_j|$

We can choose among fairness measures by checking to what extent they satisfy so-called fairness axioms.

Every fairness axiom is a property that we want our fairness measure to satisfy, and reflects some value judgment about fairness.

In the following, we discuss to what extent the range measure satisfies some popular fairness axioms.

Perfect fairness is equivalent to a perfectly equal payoff distribution:

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Range:

$$\max_i p_i - \min_i p_i = 0 \iff \max_i p_i = \min_i p_i \iff p_i = p_j \quad \forall i, j$$

The fairness measure should be indifferent to permutations of the payoff vector:

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Range: Indeed, the maximum and minimum are not affected by permutations.

Lower payoffs are (weakly) preferred:

$$\mathbf{p} \succ \mathbf{p}' \Rightarrow \phi(\mathbf{p}) \geq \phi(\mathbf{p}')$$

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Range: A simple counterexample is $\mathbf{p} = (0, 1)$, $\mathbf{p}' = (1, 1)$ with $\phi(\mathbf{p}) = 1$ and $\phi(\mathbf{p}') = 0$.

In other words: The range could be optimized by artificially increasing the workload of a solution!

Equalizing transfers from better-off to worse-off agents should never worsen the fairness measure:

If $p'_i = p_i - \epsilon$, $p'_j = p_j + \epsilon$ with $0 \leq \epsilon < p_i - p_j$, then $\phi(\mathbf{p}') \leq \phi(\mathbf{p})$

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Range: Any Pigou-Dalton transfer can only decrease the maximum and increase the minimum. For example:

■ $\mathbf{p} = (0, 0, 1), \mathbf{p}' = (0, 1/2, 1/2) \Rightarrow \phi(\mathbf{p}) = 1, \phi(\mathbf{p}') = 1/2$

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Other desirable fairness axioms include:

■ **Transitivity:**

$$\phi(\mathbf{p}) \geq \phi(\mathbf{p}'), \phi(\mathbf{p}') \geq \phi(\mathbf{p}'') \Rightarrow \phi(\mathbf{p}) \geq \phi(\mathbf{p}'')$$

■ **Scale invariance:**

$$\phi(\lambda \mathbf{p}) = \phi(\mathbf{p}) \quad \forall \lambda \neq 0$$

■ **Translation invariance:**

$$\phi(\mathbf{p} + \lambda \mathbf{1}) = \phi(\mathbf{p})$$

Property	Min-max	Lexicogr. min-max	Range	Mean abs. dev.	Standard dev.	Gini coeff.
Normalisation			•	•	•	•
Transitivity	•	•	•	•	•	•
Scale inv.						•
Translation inv.			•	•	•	•
Anonymity	•	•	•	•	•	•
Monotonicity	Weak	Strong				
Pigou-Dalton	Weak	Strong	Weak	Weak	Strong	Strong

The price of fairness measures the relative cost of obtaining a fair solution, and is defined as

$$POF = \frac{z_{FAIR}^* - z_{EFFICIENT}^*}{z_{EFFICIENT}^*}. \quad (1)$$

Typically, decision-makers are interested in the POF for different fairness levels, i.e., the Pareto curve balancing fairness and efficiency.

One way of characterizing the fairness-efficiency trade-off is through bi-objective optimization models.

Here, parameter $\alpha \in [0, 1]$ governs the trade-off between fairness $\phi(\mathbf{p}(\mathbf{x}))$ and efficiency $\mathbf{c}^\top \mathbf{x}$.

$$\min \quad \alpha \cdot \mathbf{c}^\top \mathbf{x} + (1 - \alpha) \cdot \phi(\mathbf{p}(\mathbf{x})) \quad (2a)$$

$$\text{s.t.} \quad \mathbf{x} \in \mathcal{X}. \quad (2b)$$

Varying α allows us to obtain the full Pareto-curve.

Alternatively, we can minimize the fairness measure subject to an efficiency bound:

$$\min \quad \phi(\mathbf{p}(\mathbf{x})) \tag{3a}$$

$$\text{s.t.} \quad \mathbf{c}^\top \mathbf{x} \leq (1 + \epsilon) \mathbf{z}^* \tag{3b}$$

$$\mathbf{x} \in \mathcal{X}. \tag{3c}$$

Here, $\epsilon \geq 0$ models the price of fairness and $\mathbf{z}^*$ is the cost of the most efficient solution.

Equivalently, we could minimize costs while imposing a bound on unfairness.

The following is a route-based formulation for the range fairness measure:

$$\min \quad \eta - \gamma \quad (4a)$$

$$\text{s.t.} \quad \sum_{k \in K} \sum_{r \in R} a_{ir} x_{rk} = 1 \quad \forall i \in N \setminus \{0\} \quad (4b)$$

$$\sum_{k \in K} \sum_{r \in R} c_r x_{rk} \leq (1 + \epsilon) z^* \quad (4c)$$

$$\sum_{r \in R} x_{rk} = 1 \quad \forall k \in K \quad (4d)$$

$$\gamma \leq \sum_{r \in R} p_r x_{rk} \leq \eta \quad \forall k \in K \quad (4e)$$

$$x_{rk} \in \{0, 1\} \quad \forall k \in K, r \in R. \quad (4f)$$

Variables η and γ denote the maximum and minimum payoff, respectively.

Enforcing fairness on single-shot optimization models can be quite costly, or even impossible.

Moreover, in practice many planning problems are repeated over time:

- Crew scheduling
- Vehicle routing

Hence, the relevant concept is fairness over time

- Fairness with respect to aggregated payoffs over the full planning period

The price of fairness over time is typically much lower. **Why?**

We can formulate fairness over time in a model as follows:

$$\min \quad \phi(\mathbf{p}) \tag{5a}$$

$$\text{s.t.} \quad \mathbf{p} = \frac{1}{T} \sum_{t=1}^T \mathbf{p}(\mathbf{x}_t) \tag{5b}$$

$$\sum_{t=1}^T \mathbf{c}_t^\top \mathbf{x}_t \leq (1 + \epsilon) \sum_{t=1}^T \mathbf{z}_t^* \tag{5c}$$

$$\mathbf{x}_t \in \mathcal{X}_t \quad \forall t = 1, \dots, T. \tag{5d}$$

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$$\mathbf{x}_t \in \mathcal{X}_t \quad \forall t = 1, \dots, T. \tag{5d}$$

However, this model might not be very useful in practice:

- Formulation does not scale well, and is highly symmetric
- In practice, information is revealed dynamically

At NS, drivers and guards perform work in the form of duties.

Following major strikes, NS has agreed upon a set of rules to fairly balance workload over the various crew bases.

These Sharing-Sweet-and-Sour rules apply to different duty attributes, such as:

- Average fraction of double-decker work (sour)
- Average fraction of aggression work (sour)
- Track variation (sweet)
- Unique kilometers (sweet)

However, the current rules do not guarantee fairness across crew members:

- There remains significant variation within crew bases
- In practice, the rules are not always met due to many rescheduling steps

This is why NS investigates a new approach:

- Apply rules at the individual level
- Apply rules as close as possible to the day of operation

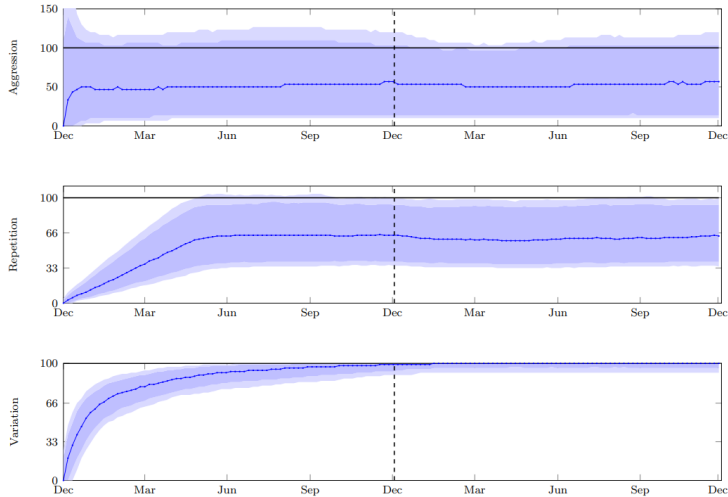
To facilitate this approach, we developed a penalty-based feedback mechanism that can be used in the crew scheduling step.

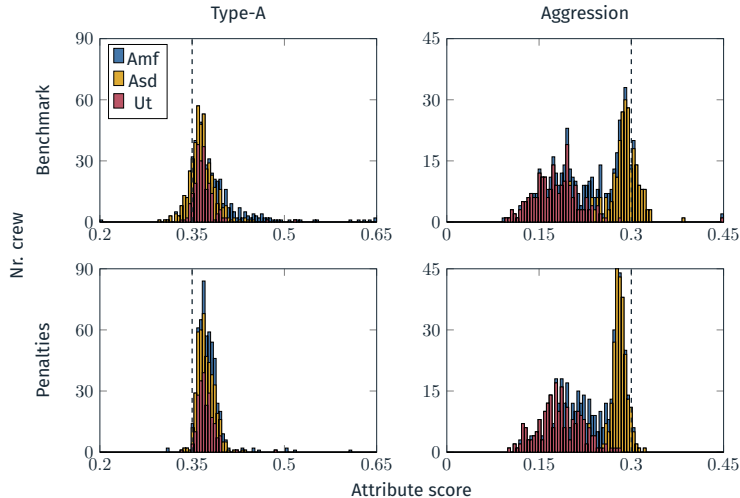
1. Keep track of crew members whose recent work history violates the SS&S rules.
2. Penalize the assignment of duties that would lead to further violation.

The penalties adjust dynamically throughout the planning period.

Selecting duties that minimize penalties leads to a balanced work allocation.

Case Study: NS





Study fairness in capacitated vehicle routing

Compare the computational performance of two route-based formulations for minimizing the range of route distances

Characterize the trade-off between fairness and efficiency

Optimization under Uncertainty

So far, we have assumed perfect information about all problem parameters.

However, many real-life parameters are uncertain:

- Not exactly known
- Measurement errors
- Subject to noise and external factors (traffic, weather)

A deterministic solution might not be feasible or cost-efficient with respect to the true parameters.

- Optimal solutions contain little slack

For example, we might solve

$$\min \quad \mathbf{c}^\top \mathbf{x} \quad (6a)$$

$$\text{s.t.} \quad A\mathbf{x} \geq \mathbf{b} \quad (6b)$$

$$\mathbf{x} \geq 0, \quad (6c)$$

whereas the true model is

$$\min \quad \mathbf{c}^\top \mathbf{x} \quad (7a)$$

$$\text{s.t.} \quad \tilde{A}\mathbf{x} \geq \tilde{\mathbf{b}} \quad (7b)$$

$$\mathbf{x} \geq 0 \quad (7c)$$

for some uncertain parameters $(\tilde{A}, \tilde{\mathbf{b}}) \neq (A, \mathbf{b})$. Since $A\mathbf{x} \geq \mathbf{b}$ does not imply $\tilde{A}\mathbf{x} \geq \tilde{\mathbf{b}}$, the solution might not be feasible anymore!

To overcome these issues, we aim to find robust solutions.

We say a solution is robust when it remains cost-efficient and feasible under (minor) deviations from the nominal parameters.

Typically, accounting for a little bit of uncertainty is already much better than doing nothing!

Public transport systems, such as railways, are subject to frequent delays. **What are possible sources of delays?**



These delays can affect the whole network through ‘cascading’:

- A single train unit or crew member traversing the country
- Bottleneck capacity at a station

To handle uncertainty, we add buffer times to the schedule and use real-time rescheduling tools.

The true customer demand can be uncertain.

Hence, a vehicle might not have sufficient capacity to cover all customers along its route.

We can either build in slack capacity, or use recourse actions such as replenishing at the depot.

In practice, there might be more sources of uncertainty. **Such as?**

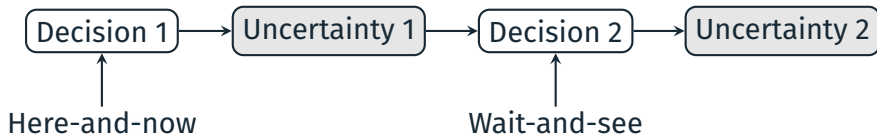
Static Robust Optimization



Static Robust Optimization



Dynamic Robust Optimization



For simplicity, we assume that the uncertain parameters are in the constraints only.

The 'true' data-generating process is often unknown or too complex to capture exactly.

Instead, we aim to approximate the parameter uncertainty in one of the following ways:

- Probability distribution
- Uncertainty set
- **Scenario set**

Scenario-based formulations offer a simple way of dealing with uncertainty:

- Assume there exists a finite number of possible parameter realizations, i.e., scenarios
- We want to choose a solution that is feasible in every scenario
- Hence, we ‘copy’ constraints for each scenario

Let S be the scenario set. The scenario-based formulation becomes:

$$\min \quad \mathbf{c}^\top \mathbf{x} \tag{8a}$$

$$\text{s.t.} \quad \mathbf{A}_s \mathbf{x} \geq \mathbf{b}_s \qquad \forall s \in S \tag{8b}$$

$$\mathbf{x} \geq 0. \tag{8c}$$

Consider the capacitated vehicle routing problem (CVRP) with uncertain demand.

- For each customer i , we only know the nominal demand q_i .
- However, upon arriving at a customer, we observe the true demand $\tilde{q}_i$.

How does this affect the standard formulation?

The standard two-index formulation for the CVRP reads:

$$\min \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ij} \quad (9a)$$

$$\text{s.t.} \quad \sum_{j \in N} x_{ij} = 1 \quad \forall i \in N \setminus \{0\} \quad (9b)$$

$$\sum_{i \in N} x_{ij} = 1 \quad \forall j \in N \setminus \{0\} \quad (9c)$$

$$u_i - u_j + Q x_{ij} \leq Q - q_j \quad \forall i \in N, j \in N \setminus \{0\}, i \neq j \quad (9d)$$

$$q_i \leq u_i \leq Q \quad \forall i \in N \setminus \{0\} \quad (9e)$$

$$u_0 = 0 \quad (9f)$$

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in N. \quad (9g)$$

Recall that variable u_i is the total required load when arriving at customer i .

We wish to come up with a scenario-based robust formulation for the CVRP.

- Select a single set of routes, that is feasible in multiple demand scenarios.

Let S be a scenario set. Every scenario $s \in S$ corresponds to a possible demand realisation.

- Denote by q_i^s the demand of customer i in scenario s

What does a scenario-based formulation look like?

The scenario-based formulation reads:

$$\min \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ij} \quad (10a)$$

$$\text{s.t.} \quad \sum_{j \in N} x_{ij} = 1 \quad \forall i \in N \setminus \{0\} \quad (10b)$$

$$\sum_{i \in N} x_{ij} = 1 \quad \forall j \in N \setminus \{0\} \quad (10c)$$

$$u_i^s - u_j^s + Q x_{ij} \leq Q - q_j^s \quad \forall s \in S, i \in N, j \in N \setminus \{0\}, i \neq j \quad (10d)$$

$$q_i^s \leq u_i^s \leq Q \quad \forall s \in S, i \in N \setminus \{0\} \quad (10e)$$

$$u_0^s = 0 \quad \forall s \in S \quad (10f)$$

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Variable u_i^s is the total required load when arriving at customer i in scenario s .

Considerations:

- A small scenario set might not be representative of the true parameter distribution
- A large scenario set might make the model computationally intractable

In practice, the following approaches are common:

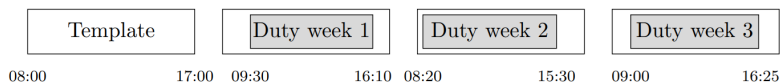
- Use a representative subset of historical data (e.g., historical timetables)
- Cutting planes
 1. Initialise model with small scenario set
 2. Iteratively add most violated scenario until some stopping criterion is met

NS wishes to communicate the rosters to crew members well in advance, so they can schedule their personal lives around work.

At the same time, NS wishes to maintain flexibility to account for changes in the timetable.

Templates offer a way out of this dilemma:

- Templates specify time windows, during which duties can later be scheduled.

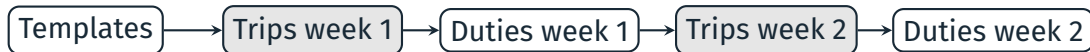


Templates must be scheduled for a planning period of several weeks.

- At this point in time, weekly timetables are not finalized yet.

Once the weekly timetable is finished, we assign duties to templates that cover trips in the timetable.

- Excess duties are available to ensure all trips are covered.



What are the here-and-now and wait-and-see decisions?

Timetable uncertainty cannot be easily captured in (linear) problem parameters:

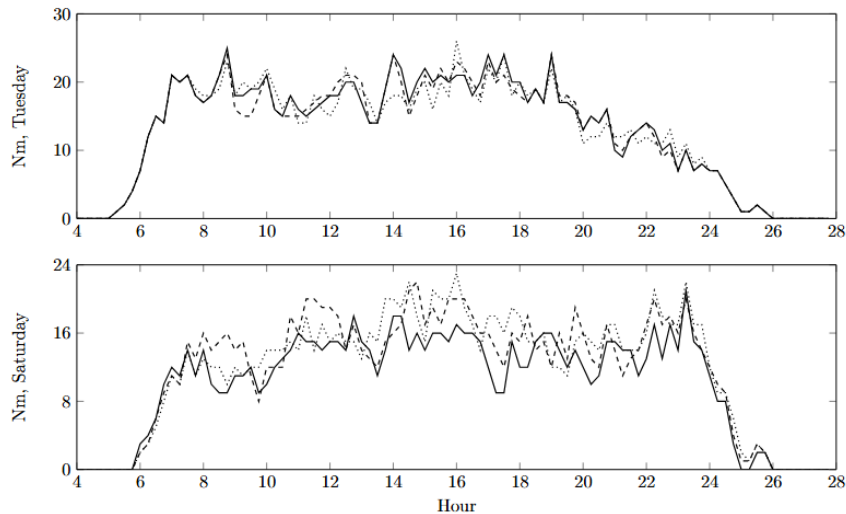
- Different train trips
- Different connections between train trips

Hence, we use a scenario-based approach:

- Every scenario contains a set of trips and connections between them

NS has historical planning data for multiple weeks, which we can use to construct scenarios.

Robust Crew Scheduling: Scenarios



We use the following notation:

- P : Set of possible templates
- S : Set of scenarios
- K_s : Set of tasks to cover in scenario $s \in S$
- Δ_p^s : Set of duties that fit template $p \in P$ in scenario $s \in S$

Define the following decision-variables:

- $y_p \in \mathbb{N}$: Number of templates of type p
- $x_p^{\delta s} \in \{0, 1\}$: Whether duty $\delta \in \Delta_p^s$ is used
- $v_p^s \geq 0$: Excess capacity of template p in scenario s
- $\eta \geq 0$: Worst-case total excess capacity across all scenarios

$$\min \quad \sum_{p \in P} c_p y_p + \eta \quad (11a)$$

$$\text{s.t.} \quad \sum_{p \in P} c_E v_p^s \leq \eta \quad \forall s \in S \quad (11b)$$

$$\sum_{\delta \in \Delta_p^s} x_p^{\delta s} \leq y_p + v_p^s \quad \forall s \in S, p \in P \quad (11c)$$

$$\sum_{p \in P} \sum_{\delta \in \Delta_p^s} a_{pk}^{\delta s} x_p^{\delta s} \geq 1 \quad \forall s \in S, k \in K_s \quad (11d)$$

$$x_p^{\delta s} \in \{0, 1\} \quad \forall s \in S, p \in P, \delta \in \Delta_p^s \quad (11e)$$

$$y_p \in \mathbb{N} \quad \forall p \in P \quad (11f)$$

$$v_p^s \geq 0 \quad \forall s \in S, p \in P. \quad (11g)$$

We evaluate the solutions on four weeks of planning data.

Scenarios	Templates	Duties	Excess	Workload (h)
1	60.18	62.14	2.50	7.72
2	61.29	61.96	1.15	7.74
3	61.69	62.06	0.90	7.73

Increasing the number of scenarios leads to higher first-stage and lower second-stage costs.

Dynamic robust optimization features two types of decisions:

- Here-and-now decisions
- Wait-and-see decisions

Such decisions can be easily incorporated in scenario-based formulations:

- Here-and-now: Scenario-independent variables
- Wait-and-see: Scenario-indexed variables

In general, solving dynamic problems to optimality requires advanced solution methods.

Recourse policies form a simple alternative to exact dynamic robust optimization.

They specify how to repair an infeasible solution after uncertainty has realised.

In the CVRP with demand uncertainty, we could do the following:

- Replenish when we arrive at a customer with insufficient remaining capacity
- Return to the depot when the vehicle capacity is below a certain threshold
- Skip a customer, and revisit later

We can assess the robustness of a solution as follows:

- Simulation, sampling parameters from some distribution
- Evaluating on historical data

Some metrics of interest are:

- Average cost
- Worst-case cost
- Fraction feasible
- Average violation
- Worst-case violation

Study the CVRP with stochastic demand

Evaluate robustness of a deterministic solution

Implement a simple scenario-based model and solve it with cutting planes

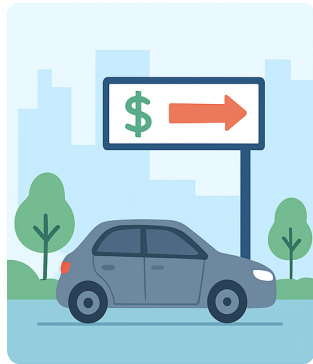
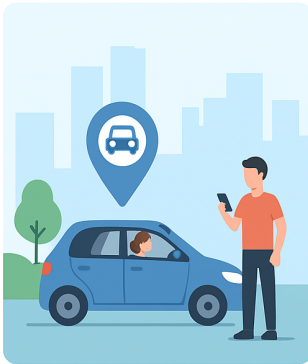
Propose and implement a simple recourse policy

Multi-Agent Optimization

So far, we studied optimization models featuring a single decision-maker.

In reality, there might be multiple agents that either collaborate with or compete against each other.

These agents may make decisions either simultaneously or sequentially.



	Cooperative	Non-cooperative
Simultaneous	Cost sharing	Nash equilibrium
Sequential	Empty?	Leader-follower

Consider a simultaneous, cooperative setting where a coalition of players jointly solves an optimization problem.

We wish to fairly allocate the costs (profits) of this collaboration.

Notation:

- N : Set of players
- $v(S)$: Cost (profit) of coalition $S \subseteq N$

Our goal is to find a value function $\phi_i(v)$ that fairly allocates costs (profits) to each player $i \in N$.

We would like a value function satisfying the following properties:

1. **Efficiency:**

$$\sum_{i \in N} \phi_i(v) = v(N)$$

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4. **Linearity:**

$$\phi_i(\mathbf{v} + \mathbf{w}) = \phi_i(\mathbf{v}) + \phi_i(\mathbf{w}), \quad \phi_i(\alpha \mathbf{v}) = \alpha \phi_i(\mathbf{v})$$

As it turns out, the Shapley value is the unique value function satisfying all these properties.

The Shapley value is defined as

$$\phi_i(\mathbf{v}) = \sum_{\underbrace{S \subseteq N \setminus \{i\}}_{\text{coalitions without } i}} \underbrace{\frac{|S|!(|N| - |S| - 1)!}{|N|!}}_{\text{weight factor}} \underbrace{(v(S \cup \{i\}) - v(S))}_{\text{marginal contribution of } i}$$

Informally, the Shapley value computes the marginal contribution of each player to all possible coalitions.

Shapley Value: Example

S	$v(S)$	$v(S \cup \{1\}) - v(S)$	$\frac{ S !(N - S -1)!}{ N !}$
$\emptyset$	0	2	2/6
$\{1\}$	2	-	1/6
$\{2\}$	3	5	1/6
$\{3\}$	1	5	1/6
$\{1, 2\}$	8	-	2/6
$\{1, 3\}$	6	-	2/6
$\{2, 3\}$	7	3	2/6
$\{1, 2, 3\}$	10	-	1/6

S	$v(S)$	$v(S \cup \{1\}) - v(S)$	$\frac{ S !(N - S -1)!}{ N !}$
$\emptyset$	0	2	2/6
$\{1\}$	2	-	1/6
$\{2\}$	3	5	1/6
$\{3\}$	1	5	1/6
$\{1, 2\}$	8	-	2/6
$\{1, 3\}$	6	-	2/6
$\{2, 3\}$	7	3	2/6
$\{1, 2, 3\}$	10	-	1/6

The Shapley value of player 1 equals

$$\phi_1(\mathbf{v}) = \frac{2}{6} \cdot 2 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 5 + \frac{2}{6} \cdot 3 = \frac{20}{6}.$$



Typically, the leader and followers have conflicting objectives.

Examples:

- **Toll setting:** Government sets tolls, drivers choose cheapest paths
- **Capacity planning:** Operator builds infrastructure, users respond with route choices
- **Ticket pricing:** Transport operator sets fares, passengers choose modes

As motivating example, consider a simple toll setting problem:

- Road network $G = (N, A)$, where arc $a \in A$ has travel time t_a
- OD-pairs K , representing road users traveling from origin to destination

First, the toll-setter (leader) sets toll prices on each road segment.

Next, all road users (followers) determine their cheapest paths.

The leader specifies a toll T_a for each arc a . The goal is to maximise total revenue.

We can formulate the leader's problem as:

$$\max_T \sum_{a \in A} T_a f_a \quad (12a)$$

$$\text{s.t. } l_a \leq T_a \leq u_a \quad \forall a \in A. \quad (12b)$$

The flow f is determined by the followers in the lower-level problem.

Next, all users k choose flows f_a^k to travel from origin to destination. Their goal is to minimize the sum of travel time and toll costs.

We can formulate the followers' problem as:

$$\min_f \sum_{k \in K} \sum_{a \in A} (t_a + T_a) f_a^k \quad (13a)$$

$$\text{s.t.} \quad \sum_{a \in A^-(i)} f_a^k - \sum_{a \in A^+(i)} f_a^k = d_i^k \quad \forall i \in N, k \in K \quad (13b)$$

$$f_a^k \geq 0 \quad \forall a \in A, k \in K. \quad (13c)$$

A more realistic model would incorporate road congestion here.

The full bi-level model reads:

$$\max_T \sum_{a \in A} T_a f_a \quad (14a)$$

$$\text{s.t. } l_a \leq T_a \leq u_a \quad \forall a \in A \quad (14b)$$

$$f \in \arg \min_f \sum_{k \in K} \sum_{a \in A} (t_a + T_a) f_a^k \quad (14c)$$

$$\text{s.t. } \sum_{a \in A^-(i)} f_a^k - \sum_{a \in A^+(i)} f_a^k = d_i^k \quad \forall i \in N, k \in K \quad (14d)$$

$$f_a^k \geq 0 \quad \forall a \in A. \quad (14e)$$

Our toll setting model omits some important practical considerations, like congestion effects.

Nonetheless, it already gives rise to a complicated, bi-level optimization model.

Solving these models to optimality is challenging, and outside the scope of this course.

System optimum: Leader centrally minimizes aggregate objective function

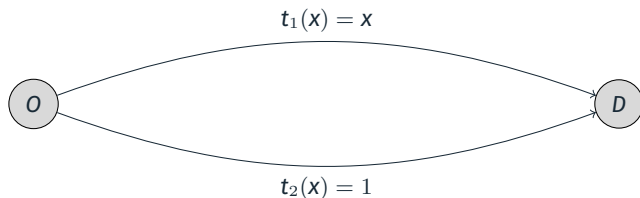
User equilibrium: All followers minimize their own objective function (subject to actions by others)

Price of anarchy:

$$POA = \frac{z_{USER}^* - z_{SYSTEM}^*}{z_{SYSTEM}^*}. \quad (15)$$

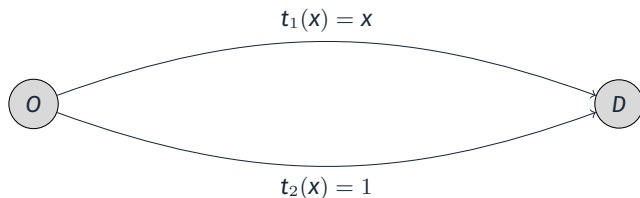
Informally, the PoA measures the welfare loss due to selfish behaviour.

Consider a simple traffic model with congestion:



The total travel time equals $t(x) = x \cdot t_1(x) + (1 - x) \cdot t_2(1 - x) = x^2 + (1 - x)$.

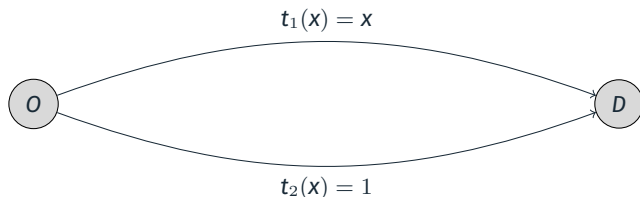
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System-optimum: $x = \frac{1}{2}$ with total travel time $t(\frac{1}{2}) = \frac{3}{4}$

User-optimum: $x = 1$ with total travel time $t(1) = 1$

A set of travelers wish to travel from their origin to their destination through a road network.

Their vehicles have a limited driving range, and must recharge before the range is depleted.

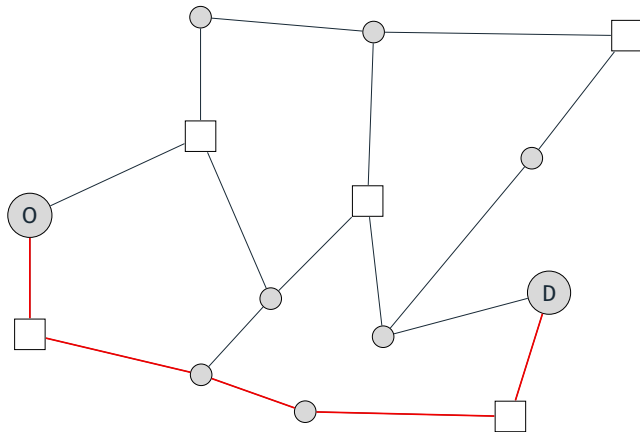
- Vehicles start their trip with full charge.

We can locate charging facilities at which travelers can recharge.

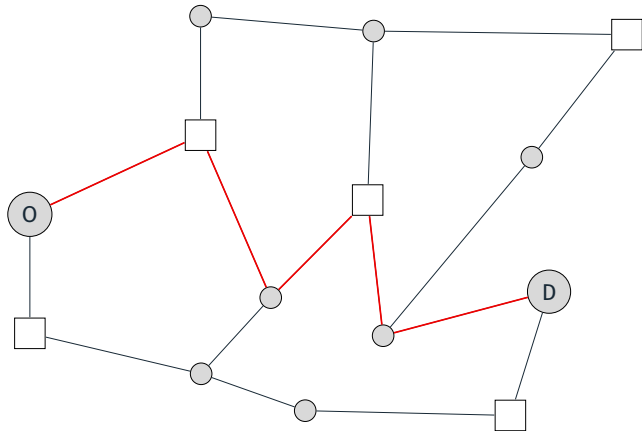
What is the minimum number of facilities we must locate to allow all travelers to reach their destination without running out of charge?

Charging Facility Location: Example

Suppose that R equals two arcs. This path is not charge-feasible:



But this path is:



Let K be the set of origin-destination pairs. Every $k \in K$ has volume V_k , origin $O_k \in N$, and destination $D_k \in N$.

We first construct an auxiliary range-network $G = (N, A)$:

- A only contains arc (i, j) when j is within driving range R from i .

Next, introduce the following decision-variables:

- $y_i \in \{0, 1\}$: Open a facility at location i or not
- $x_{ij}^k \geq 0$: Flow of pair k traveling along arc (i, j)

This formulation allows splittable flow. **How could we avoid this?**

A formulation of the CFL now reads:

$$\min \sum_{i \in N} y_i \quad (16a)$$

$$\text{s.t.} \quad \sum_{j: (i,j) \in A} x_{ij}^k - \sum_{j: (j,i) \in A} x_{ji}^k = d_i^k \quad \forall i \in N, k \in K \quad (16b)$$

$$x_{ij}^k \leq y_j \quad \forall k \in K, (i,j) \in A : j \neq D_k \quad (16c)$$

$$x_{ij}^k \geq 0 \quad \forall (i,j) \in A, k \in K \quad (16d)$$

$$y_i \in \{0, 1\} \quad \forall i \in N. \quad (16e)$$

Is this a leader-follower problem?

Implement the (capacitated) charging facility location model

Compute the gains of collaboration across multiple companies

Fairly allocate costs using the Shapley value

Evaluate the effect of selfish user routing